Activity_Course 2 TikTok project lab

April 14, 2025

1 TikTok Project

Course 2 - Get Started with Python

Welcome to the TikTok Project!

You have just started as a data professional at TikTok.

The team is still in the early stages of the project. You have received notice that TikTok's leadership team has approved the project proposal. To gain clear insights to prepare for a claims classification model, TikTok's provided data must be examined to begin the process of exploratory data analysis (EDA).

A notebook was structured and prepared to help you in this project. Please complete the following questions.

2 Course 2 End-of-course project: Inspect and analyze data

In this activity, you will examine data provided and prepare it for analysis.

The purpose of this project is to investigate and understand the data provided. This activity will:

1. Acquaint you with the data
2. Compile summary information about the data
3. Begin the process of EDA and reveal insights contained in the data
4. Prepare you for more in-depth EDA, hypothesis testing, and statistical analysis

The goal is to construct a dataframe in Python, perform a cursory inspection of the provided dataset, and inform TikTok data team members of your findings. *This activity has three parts:*

Part 1: Understand the situation * How can you best prepare to understand and organize the provided TikTok information?

Part 2: Understand the data

- Create a pandas dataframe for data learning and future exploratory data analysis (EDA) and statistical activities
- Compile summary information about the data to inform next steps

Part 3: Understand the variables

- Use insights from your examination of the summary data to guide deeper investigation into variables

To complete the activity, follow the instructions and answer the questions below. Then, you will us your responses to these questions and the questions included in the Course 2 PACE Strategy Document to create an executive summary.

Be sure to complete this activity before moving on to Course 3. You can assess your work by comparing the results to a completed exemplar after completing the end-of-course project.

3 Identify data types and compile summary information

Throughout these project notebooks, you'll see references to the problem-solving framework PACE. The following notebook components are labeled with the respective PACE stage: Plan, Analyze, Construct, and Execute.

4 PACE stages

- `[Plan](#scrollTo=psz51YkZVwtN&line=3&uniqifier=1)`
- `[Analyze](#scrollTo=mA7Mz_SnI8km&line=4&uniqifier=1)`
- `[Construct](#scrollTo=Lca9c8XON8lc&line=2&uniqifier=1)`
- `[Execute](#scrollTo=401PgchTPr4E&line=2&uniqifier=1)`

4.1 PACE: Plan

Consider the questions in your PACE Strategy Document and those below to craft your response:

4.1.1 Task 1. Understand the situation

- How can you best prepare to understand and organize the provided information?

Begin by exploring your dataset and consider reviewing the Data Dictionary.

The TikTok leadership team has provided me with the data. I plan to prepare the data and start to explore some of its key attributes to make sense of the data and share my findings with the data science manager (Rosie Mae Bradshaw). Firstly, I will import the necessary packages and start by loading the data into the jupyter notebook environment to create a data frame. Next, I will begin to identify variables within the data that will be used later for data learning and exploratory data analysis (EDA). Lastly, I will summarise and report my findings to the data science manager (Rosie Mae Bradshaw).

4.2 PACE: Analyze

Consider the questions in your PACE Strategy Document to reflect on the Analyze stage.

4.2.1 Task 2a. Imports and data loading

Start by importing the packages that you will need to load and explore the dataset. Make sure to use the following import statements: * `import pandas as pd`

- `import numpy as np`

[2]:
```
# Import packages
import pandas as pd
import numpy as np
```

Then, load the dataset into a dataframe. Creating a dataframe will help you conduct data manipulation, exploratory data analysis (EDA), and statistical activities.

Note: As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

[3]:
```
# Load dataset into dataframe
data = pd.read_csv("tiktok_dataset.csv")
```

4.2.2 Task 2b. Understand the data - Inspect the data

View and inspect summary information about the dataframe by **coding the following:**

1. `data.head(10)`
2. `data.info()`
3. `data.describe()`

Consider the following questions:

Question 1: When reviewing the first few rows of the dataframe, what do you observe about the data? What does each row represent?

Question 2: When reviewing the `data.info()` output, what do you notice about the different variables? Are there any null values? Are all of the variables numeric? Does anything else stand out?

Question 3: When reviewing the `data.describe()` output, what do you notice about the distributions of each variable? Are there any questionable values? Does it seem that there are outlier values?

Response:

Question 1: Each row of the data frame represents a video it outlines information such as the claim status, video duration, verified and author ban status, number of likes, shares and comments.

Question 2: The data frame contains 19382 entries and some columns contain null values. Some of the columns contain exactly the same amount of null values 298. This might indicate that missing data might be related across columns.

Question 3: The mean amount for video_view_count is significantly more than the median (50% range) indicating that the data is heavily skewed to the right and that a few videos received a significant amount of viewing attention. Video_duration spans from 5 seconds to 60 seconds suggesting that the videos are all short from the TikTok platform. Minimum view_count indicates that only videos with a minimum of 20 views are included in the data set.

[3]:
```python
# Display and examine the first ten rows of the dataframe
data.head(10)
```

[3]:
```
    # claim_status    video_id  video_duration_sec  \
0   1        claim  7017666017                  59
1   2        claim  4014381136                  32
2   3        claim  9859838091                  31
3   4        claim  1866847991                  25
4   5        claim  7105231098                  19
5   6        claim  8972200955                  35
6   7        claim  4958886992                  16
7   8        claim  2270982263                  41
8   9        claim  5235769692                  50
9  10        claim  4660861094                  45

                            video_transcription_text verified_status  \
0  someone shared with me that drone deliveries a…    not verified
1  someone shared with me that there are more mic…    not verified
2  someone shared with me that american industria…    not verified
3  someone shared with me that the metro of st. p…    not verified
4  someone shared with me that the number of busi…    not verified
5  someone shared with me that gross domestic pro…    not verified
6  someone shared with me that elvis presley has …    not verified
7  someone shared with me that the best selling s…    not verified
8  someone shared with me that about half of the …    not verified
9  someone shared with me that it would take a 50…        verified

  author_ban_status  video_view_count  video_like_count  video_share_count  \
0      under review          343296.0           19425.0              241.0
1            active          140877.0           77355.0            19034.0
2            active          902185.0           97690.0             2858.0
3            active          437506.0          239954.0            34812.0
4            active           56167.0           34987.0             4110.0
5      under review          336647.0          175546.0            62303.0
6            active          750345.0          486192.0           193911.0
7            active          547532.0            1072.0               50.0
8            active           24819.0           10160.0             1050.0
9            active          931587.0          171051.0            67739.0
```

```
   video_download_count  video_comment_count
0                   1.0                  0.0
1                1161.0                684.0
2                 833.0                329.0
3                1234.0                584.0
4                 547.0                152.0
5                4293.0               1857.0
6                8616.0               5446.0
7                  22.0                 11.0
8                  53.0                 27.0
9                4104.0               2540.0
```

[4]:
```python
# Get summary info
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19382 entries, 0 to 19381
Data columns (total 12 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   #                         19382 non-null  int64
 1   claim_status              19084 non-null  object
 2   video_id                  19382 non-null  int64
 3   video_duration_sec        19382 non-null  int64
 4   video_transcription_text  19084 non-null  object
 5   verified_status           19382 non-null  object
 6   author_ban_status         19382 non-null  object
 7   video_view_count          19084 non-null  float64
 8   video_like_count          19084 non-null  float64
 9   video_share_count         19084 non-null  float64
 10  video_download_count      19084 non-null  float64
 11  video_comment_count       19084 non-null  float64
dtypes: float64(5), int64(3), object(4)
memory usage: 1.8+ MB
```

[5]:
```python
# Get summary statistics
data.describe()
```

[5]:
```
                  #      video_id  video_duration_sec  video_view_count  \
count  19382.000000  1.938200e+04        19382.000000      19084.000000
mean    9691.500000  5.627454e+09           32.421732     254708.558688
std     5595.245794  2.536440e+09           16.229967     322893.280814
min        1.000000  1.234959e+09            5.000000         20.000000
25%     4846.250000  3.430417e+09           18.000000       4942.500000
50%     9691.500000  5.618664e+09           32.000000       9954.500000
75%    14536.750000  7.843960e+09           47.000000     504327.000000
```

```
max    19382.000000  9.999873e+09           60.000000     999817.000000

       video_like_count  video_share_count  video_download_count  \
count       19084.000000       19084.000000          19084.000000
mean        84304.636030       16735.248323           1049.429627
std        133420.546814       32036.174350           2004.299894
min             0.000000           0.000000              0.000000
25%           810.750000         115.000000              7.000000
50%          3403.500000         717.000000             46.000000
75%        125020.000000       18222.000000           1156.250000
max        657830.000000      256130.000000          14994.000000

       video_comment_count
count         19084.000000
mean            349.312146
std             799.638865
min               0.000000
25%               1.000000
50%               9.000000
75%             292.000000
max            9599.000000
```

4.2.3 Task 2c. Understand the data - Investigate the variables

In this phase, you will begin to investigate the variables more closely to better understand them.

You know from the project proposal that the ultimate objective is to use machine learning to classify videos as either claims or opinions. A good first step towards understanding the data might therefore be examining the `claim_status` variable. Begin by determining how many videos there are for each different claim status.

[12]:
```python
# What are the different values for claim status and how many of each are in
 ↪the data?

claim_status_counts = data['claim_status'].value_counts(dropna=False)

print('Value counts for "claim_status":')
print(claim_status_counts)
```

```
Value counts for "claim_status":
claim      9608
opinion    9476
NaN         298
Name: claim_status, dtype: int64
```

Question: What do you notice about the values shown?

Response:

There are a total of 9608 claims, 9476 opinions and 298 NaN values under the claim_status variable.

Next, examine the engagement trends associated with each different claim status.

Start by using Boolean masking to filter the data according to claim status, then calculate the mean and median view counts for each claim status.

[18]:
```
# What is the average view count of videos with "claim" status?
claims = data[data['claim_status']== 'claim']

print('Mean view count claims:', claims['video_view_count'].mean())
print('Median view count claims:',claims['video_view_count'].median())
```

```
Mean view count claims: 501029.4527477102
Median view count claims: 501555.0
```

[19]:
```
# What is the average view count of videos with "opinion" status?

claims = data[data['claim_status']=='opinion']

print('Mean view opinion count:', claims['video_view_count'].mean())
print('Median view opinion count:', claims['video_view_count'].mean())
```

```
Mean view opinion count: 4956.43224989447
Median view opinion count: 4956.43224989447
```

Question: What do you notice about the mean and media within each claim category?

Response: Mean and medium amounts for claims are close to the same amount but there is a 526 difference in values. Opinions however have exactly the same mean and median amount.

Now, examine trends associated with the ban status of the author.

Use `groupby()` to calculate how many videos there are for each combination of categories of claim status and author ban status.

[20]:
```
# Get counts for each group combination of claim status and author ban status

data.groupby(['claim_status','author_ban_status']).count()[['#']]
```

[20]:

claim_status	author_ban_status	#
claim	active	6566
	banned	1439
	under review	1603
opinion	active	8817
	banned	196
	under review	463

Question: What do you notice about the share count of banned authors, compared to that of active authors? Explore this in more depth.

Use groupby() to group the data by author_ban_status, then use agg() to get the count, mean, and median of each of the following columns: * video_view_count * video_like_count * video_share_count

Remember, the argument for the agg() function is a dictionary whose keys are columns. The values for each column are a list of the calculations you want to perform.

Question: What do you notice about the number of claims videos with banned authors? Why might this relationship occur?

Response: This relationship occurs because a claim made by an author can be false or misleading and can have detrimental consequences. Therefore an author is more likely to get banned from the platform.

Continue investigating engagement levels, now focusing on author_ban_status.

Calculate the median video share count of each author ban status.

[15]:
```
data.groupby(['author_ban_status']).agg(
    {'video_view_count': ['mean', 'median'],
     'video_like_count': ['mean', 'median'],
     'video_share_count': ['mean', 'median']})
```

[15]:
```
                  video_view_count           video_like_count           \
                              mean    median             mean    median
author_ban_status
active               215927.039524    8616.0     71036.533836    2222.0
banned               445845.439144  448201.0    153017.236697  105573.0
under review         392204.836399  365245.5    128718.050339   71204.5

                  video_share_count
                               mean   median
author_ban_status
active                 14111.466164    437.0
banned                 29998.942508  14468.0
under review           25774.696999   9444.0
```

[14]:
```
# What's the median video share count of each author ban status?
data.groupby(['author_ban_status']).
 ↪median(numeric_only=True)[['video_share_count']]
```

[14]:
```
                   video_share_count
author_ban_status
active                         437.0
banned                       14468.0
under review                  9444.0
```

[13]:
```
data.groupby(['author_ban_status']).agg(
    {'video_view_count': ['count', 'mean', 'median'],
     'video_like_count': ['count', 'mean', 'median'],
```

```
    'video_share_count': ['count', 'mean', 'median']
    })
```

[13]:

```
                 video_view_count                         video_like_count  \
                            count          mean    median            count
author_ban_status
active                      15383 215927.039524    8616.0            15383
banned                       1635 445845.439144  448201.0             1635
under review                 2066 392204.836399  365245.5             2066

                                          video_share_count                \
                          mean    median              count          mean
author_ban_status
active             71036.533836    2222.0             15383  14111.466164
banned            153017.236697  105573.0              1635  29998.942508
under review      128718.050339   71204.5              2066  25774.696999


                    median
author_ban_status
active               437.0
banned             14468.0
under review        9444.0
```

Question: What do you notice about the number of views, likes, and shares for banned authors compared to active authors?

Response: The number of views, likes and shares shared for content from banned users are noticeably higher than for active users thus indicating that the claims create a response in viewers that could have a significant negative impact in general.

Now, create three new columns to help better understand engagement rates: * `likes_per_view`: represents the number of likes divided by the number of views for each video * `comments_per_view`: represents the number of comments divided by the number of views for each video * `shares_per_view`: represents the number of shares divided by the number of views for each video

[16]:

```
# Create a likes_per_view column
data['likes_per_view'] = data['video_like_count']/ data['video_view_count']

# Create a comments_per_view column
data['comments_per_view'] = data['video_comment_count']/␣
 ↪data['video_view_count']

# Create a shares_per_view column
data['shares_per_view'] = data['video_share_count']/ data['video_view_count']
```

Use `groupby()` to compile the information in each of the three newly created columns for each combination of categories of claim status and author ban status, then use `agg()` to calculate the count, the mean, and the median of each group.

[18]:
```
data.groupby(['claim_status', 'author_ban_status']).agg(
    {'likes_per_view': ['count', 'mean', 'median'],
     'comments_per_view': ['count', 'mean', 'median'],
     'shares_per_view': ['count', 'mean', 'median']})
```

[18]:
```
                               likes_per_view                      \
                                        count      mean    median
claim_status author_ban_status
claim        active                      6566  0.329542  0.326538
             banned                      1439  0.345071  0.358909
             under review                1603  0.327997  0.320867
opinion      active                      8817  0.219744  0.218330
             banned                       196  0.206868  0.198483
             under review                 463  0.226394  0.228051

                               comments_per_view                      \
                                           count      mean    median
claim_status author_ban_status
claim        active                         6566  0.001393  0.000776
             banned                         1439  0.001377  0.000746
             under review                   1603  0.001367  0.000789
opinion      active                         8817  0.000517  0.000252
             banned                          196  0.000434  0.000193
             under review                    463  0.000536  0.000293

                               shares_per_view
                                         count      mean    median
claim_status author_ban_status
claim        active                       6566  0.065456  0.049279
             banned                       1439  0.067893  0.051606
             under review                 1603  0.065733  0.049967
opinion      active                       8817  0.043729  0.032405
             banned                        196  0.040531  0.030728
             under review                  463  0.044472  0.035027
```

Question:

How does the data for claim videos and opinion videos compare or differ? Consider views, comments, likes, and shares.

Response: When comparing the data there are some noticeable differences. First, opinion videos received more likes, comments and views than claim videos. Secondly, the claim videos have much higher banned and under review values. The reason for this is likely because videos that state a claim are more likely to be under review and eventually receive a banned status.

4.3 PACE: Construct

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.

4.4 PACE: Execute

Consider the questions in your PACE Strategy Document and those below to craft your response.

4.4.1 Given your efforts, what can you summarize for Rosie Mae Bradshaw and the TikTok data team?

Note for Learners: Your answer should address TikTok's request for a summary that covers the following points:

- What percentage of the data is comprised of claims and what percentage is comprised of opinions?
- What factors correlate with a video's claim status?
- What factors correlate with a video's engagement level?

Response:

- Out of 19382 videos just under 50% of them have claim status - 9608 of them.
- Engagement correlates heavily with claim status. This can be further investigated.
- Authors with banned staus have a significantly higher engagement rate than active authors. Authors with under review status fall between these two categories in terms of engagement rate.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.